

Summary: Framework Shift in Cybersecurity

The paper challenges traditional, Eurocentric cybersecurity paradigms that view human risk strictly through individual errors or organizational compliance (micro/meso levels). By introducing an extended version of **Michel Foucault's power/knowledge theory**, the authors analyze cybersecurity at a **macrosocial and societal level**.

Through a hybrid approach combining a bibliometric analysis of 910 Web of Science publications (2000–2024) and an abductive thematic review of 100 highly cited papers, the study uncovers a distinct transition: **the academic field is shifting from hard power manifestations (cybercrime, terrorism) to soft power manifestations (persuasion, algorithmic agency) and alternative forms of "counter-knowledge" (Global South, intersectional/marginalized identities)**.

1. Methodology & Data Landscape

- **Dataset & Screening:** 910 final publications across social sciences and humanities indexed in the Web of Science core collection.
- **Core Metrics:** Publication volume peaked during 2020–2021 (attributed to geopolitical events like Russian cyberwarfare and the COVID-19 pandemic) and has since plateaued.
- **Top Academic Outlets & Authors:** * *By Citation/Influence:* **MIS Quarterly**, **Deviant Behavior**, and **Computers in Human Behavior** dominate the landscape.
 - *Prolific Scholars:* Thomas Holt, Eric Leukfeldt, and Adam Bossler are the most heavily co-cited researchers.
- **Geographic Stratification:** Research is heavily centralized in the United States and the United Kingdom. This concentration provides empirical backing for the **stratification model**, demonstrating that historically and socio-economically advantaged countries retain systemic privileges in digital economy mapping and cybersecurity visibility.

2. Longitudinal Evolution: Key Trends

A time-zone analysis reveals how core conceptual terminology has evolved over the past decade:

- **2013–2016 (Punitive & Institutional Foundations):** Focus on institutional structures, identity theft, piracy, globalization, and cyberterrorism.
- **2017–2019 (Ethical Plateaus & Social Science Integrations):** Emergence of routine activity theory, planned behavior, and basic governance. *Crucially, unlike the field of Artificial Intelligence where ethics is a rapidly surging priority, cybersecurity's academic focus on direct ethical consequences peaked around 2018 and is no longer an active driver of new research.*
- **2020–2022+ (The Functional & Emerging Frontier):** Focus redirected toward defensive policy compliance, user performance, data privacy, and "exotic" virtual domains like the **metaverse** and **cryptocurrency**.

3. The Foucauldian Power/Knowledge Typology

The paper maps cybersecurity relations into a structured dual framework of Power and Knowledge:

A. Power-Oriented Dimensions

- **Hard Power (Direct Violence/Coercion):** Characterized by explicit technical acts like distributed denial of service (DDoS) attacks, state-sponsored sabotage, botnets, malware deployment, online hate crimes, and disruptive ransomware. These seek immediate operational or economic results through computational coercion.
- **Soft Power (Subtle Discipline/Influence):** Characterized by human-machine agency negotiations, trust architecture, algorithmic persuasion, social media coordination, and zero-day vulnerability dilemmas.

B. Knowledge-Oriented Dimensions ("Counter-Knowledge")

The authors highlight emergent "counter-knowledges" that actively challenge hegemonic, state-centric, or corporate narratives:

- **Global South Contexts:** Shifting away from Eurocentric models to explore how regional vulnerabilities intersect with local socio-technical realities (e.g., studies on cybercrime escalation in Nigeria, or virtual interaction risks inside conservative cultural spaces like Saudi Arabia and Egypt).
- **Ethically-Oriented Knowledge:** The operationalization of moral behavior, most notably via **ethical hacking** as an informal regulatory mechanism to expose systemic business vulnerabilities before they can be maliciously exploited.
- **Ideologically-Oriented Knowledge:** Centered on intersectional structural victims within cyberspace—specifically focusing on digital violence, workplace gender gaps, and forms of online gender-based or minority discrimination.

4. Key Legal, Policy, and AI Issues for the Department

This paper surfaces several specific touchpoints relevant to legal issues in AI, policy formulation, and digital rights:

- **The Deceptive Expansion of Surveillance:** The framework notes a rising trend where both state and non-state actors leverage the pretext of "digital safety" to push legislative initiatives that normalize intrusive surveillance techniques and dilute individual privacy rights.
- **Corporate Data Enclosure as a Risk:** Tech corporations convert personal data footprints into profitable units, fundamentally creating new vectors for digital identity theft, systemic algorithmic exposure, and privacy violations.
- **The Regulation of Informational Agents:** Highlighting research (such as Fouad, 2022) into the legal and philosophical status of "non-anthropocentric informational agents"—evaluating how autonomous code and software systems negotiate agency and challenge human/legal control.
- **Socio-Demographic AI Biases:** The paper explicitly points to systemic biases embedded in automated systems—such as the disproportionate, error-prone identification of minority groups by facial recognition models—as a vital arena where legal protections must be modernized.
- **Sanctioning Ambiguity:** The text emphasizes the legal and cultural gray zones that arise when regulating behaviors like ethical hacking or anonymous activism (e.g., the *Anonymous* collective), where actions frequently straddle the border between systemic defense and conceptual deviance.